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Digital health and Ayurveda: Real-world best practices and applications

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ABSTRACT:

India has made significant strides through initiatives such as eSanjeevani, Ayushman Bharat Digital Mission (ABDM), and the National Telemedicine Guidelines, aligning with global standards set by the World Health Organization (WHO) and the International Telecommunication Union (ITU) for accessible telehealth. Within Ayush, digitalization has taken shape through platforms such as Ayush Health Management Information System (A-HMIS), National Ayush Morbidity And Standardized Terminologies Electronic (NAMASTE) Portal, e-Aushadhi, and research repositories such as Traditional Knowledge Digital Library (TKDL) and Ayush Research Portal (ARP). The COVID-19 pandemic catalyzed telehealth adoption, spurring innovations such as tele-yoga and AyushQure, demonstrating the feasibility of integrative digital care. However, challenges persist in data interoperability, infrastructure, and ethical data governance, despite regulatory efforts like the Digital Personal Data Protection Act (2023). Future growth hinges on creating interoperable, Artificial Intelligence (AI)-ready datasets through ABDM-compliant research repositories, promoting indigenous innovation, and encouraging mobile-based Ayush services to enhance community engagement, research translation, and personalized integrative health care. This review explores the evolving landscape of digital health and its integration with Ayush systems in India, highlighting policies, platforms, and future potential. Digital health encompasses a wide array of technologies, including Telemedicine, AI, Machine Learning (ML), big data analytics, Internet of Things (IoT), and wearable devices, which together aim to improve accessibility and efficiency in healthcare delivery. This review offers a critical analysis of the integration of digital technology in Ayurveda and suggests a plan for establishing a data-driven digital Ayurveda ecosystem.

Keywords:

Artificial Intelligence, Ayurveda, Digital health, Telehealth

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INTRODUCTION

Digital health refers to an umbrella term encompassing the application of digital technologies in the field of health care.[1] Several intermeshed terms have been used under its foray, including eHealth, mHealth, telehealth, uHealth, and Healthcare 4.0.[2] The concept of digital health has pervaded all dimensions of health care and now includes telemedicine, analysis and utilization of big data, comprehensive health record digitization, Internet of Things (IoT), wireless and 5G mobile technologies, blockchain, Artificial Intelligence (AI), and Machine Learning (ML), including deep learning and wearable monitors. The use of AI can tangentially expand into the formulation of clinical support decisions using the data obtained from the wearable monitors, as well as from the medical image analysis.[1]

The Government of India (GOI) has also undertaken several steps to promote the field of telehealth: Most notably in the form of its flagship free-to-use telemedicine

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service – eSanjeevani in 2020,[3,4] and the launch of Ayushman Bharat Digital Mission (ABDM) in 2021 and telemedicine guidelines in 2020.[5] Considering the potential impact and the reach of telehealth, the World Health Organization (WHO) and the International Telecommunication Union (ITU) launched "WHO-ITU global standards for accessibility of Telehealth services" in 2022. The telehealth platforms must comply with the list of technical requirements and the standard mandates to ensure accessible telehealth service provision.[6]

While the standards and the guidelines for development of telehealth services have been laid down by global and national regulating bodies, several barriers, including infrastructure scarcity, data hosting capacity, and a lack of skilled workforce, require navigation, particularly in low-resource settings.[7] Digital health has also touched the sphere of Ayurveda, with primary attempts being made in the field of telemedicine.[8,9] This review aims to discuss the best practices and real-world models relevant to Integrative and Ayush-linked telehealth.

Real-world models

While the concept of telemedicine has been floating since the early 20th century, it was in 2001 that the first telemedicine pilot project was initiated by the Indian Space Research Organization comprising linking two hospitals of Chennai and Andhra Pradesh. Initial government steps on collection and analysis of big data include the Integrated Disease Surveillance Project, National Cancer Network (ONCONET), National Rural Telemedicine Network, National Medical College Network, and the digital medical library network.[10] Ayush digital initiatives have primarily been segregated into health information systems, research database/library, academic, and Information, Education and Communication (IEC). A few of the identified ventures include teleconsultation platforms - Ayush Health Management Information System (A-HMIS), National Ayush Morbidity And Standardized Terminologies Electronic (NAMASTE) portal, Ayush Suraksha portal, e-Aushadhi, e-Charak, Triskandha Kosha, and mobile health applications such as Siddha Initiative for Documentation of Drug Adverse Reaction (SiddAR) App. Research databases include Traditional Knowledge Digital Library (TKDL), Ayush Research Portal (ARP), Digital Helpline for Ayurveda Research Articles (DHARA), e-CHLAS, Research Management Information System (RMIS), e-Granthasamuccaya, and Ayush Sanjivani app, whereas IEC ventures include Siddha–NIS app, Yoga locator, Naturopathy-NIN app, Ayurveda e-learning, and Ayurvedic Inheritance of India.[11]

COVID-19 further steamrolled the global need for telemedicine services and in its guise, several leveraged digital technology platforms in the form of voice, audio, text, and digital data exchange to diagnose, prescribe, and follow up the patients were explored.[8,12] Studies have reported providing telemedicine-based services via video calling and messenger tool services to provide Ayurvedic consultations during critical lockdown periods. The authors also mentioned the limitations regarding such modus operandi, including the inability to perform physical examination, dependency on the adequacy of required equipment (smartphones, internet availability), and the legality of issued prescriptions.[8]

Another component of Ayush, which got a boost during COVID-19, was the emergence of tele-yoga.[13] Yoga posture sensors and/or wearable devices are counted as smart yoga devices, which serve to form a feedback mechanism and collect healthcare data.[14-16] A recent multisite tele-yoga implementation study has listed the learnings and experiences of veteran yoga teachers and has recommended expansion and incorporation of tele-yoga across other healthcare platforms and services.[17]

mHealth is an integral branch of digital health which has permeated the field of Ayurveda in recent times. An exemplar would be AyushQure: A mobile application launched by the Ayush department, Government of Madhya Pradesh. It serves to provide teleconsultation to the rural population from Ayurveda, Yoga, Unani, Siddha, and Homeopathy (Ayush) experts across the state.[18] Siddha pharmacovigilance has also made use of mHealth outreach in the form of mobile application: SiddAR. It allows real-time documentation and reporting of adverse drug reactions.[19]

Researchers attempted to increase awareness regarding herbal and Ayurvedic medicine for the cure of human and environmental health at remote sites located at higher elevations of 1034 m (3392 ft)–1794 m (5886 ft) above sea level in the Himalayan range via a ubiquitous telemedicine platform for herbal medicine.[9] The results indicated that nonsacchariferous super sweet plants provided protection to human beings from the danger of ultraviolet rays for maintaining better health at the high elevation sites. The integration of such plants could enhance accessibility, cost-effectiveness, personalized care, and reshaping of eco-friendly environment.[9]

Other real-world initiatives such as e-Sanjeevani have already made an impact in terms of outreach and impact within the overburdening population of India. As of December 2024, it has helped 330 million patients across the nation and boasts of more than 1 lakh health facilities serving as spokes, supported by 16,849 hubs and 681 online outpatient departments.[20]

The digital health sector in India largely revolves around the National Health Policy 2017 and ABDM principles to ensure inclusivity, fairness, and transparency.[21] With the goal to enable interoperability of health data within the health ecosystem and create longitudinal Electronic Health Records (EHRs) of every citizen, key registries such as the Ayushman Bharat Health Account (ABHA), Healthcare Professional Registry (HPR), Health Facility Registry (HFR), and drug registry have been launched under the ambit of ABDM.[22] ABDM is also in tandem with the Findable, Accessible, Interoperable, Reusable (FAIR) principles - advocated by the international community to promote data sharing and data reuse, and to prevent data decay.[23]

Digital health care and AI require interoperable systems and data [Figure 1]. Interoperability refers to the ability of two or more systems to exchange and utilize information, including foundational, structural, semantic, and organizational levels.[24] In this regard, Fast Healthcare Interoperability Resources (FHIR) implementation guideline for ABDM has also been developed by the National Health Authority of India.[25] FHIR is a standard for structural interoperability. Semantic interoperability enables clear data exchange. Systemized Nomenclature of Medicine-Clinical Terms (SNOMED-CT) - India Ayush extension,[26] International Classification of Disease (ICD-11) traditional medicine codes, and the NAMASTE portal[27] can ensure semantic interoperability of Ayush digital healthcare systems.

DISCUSSION

India's National Strategy for AI strives to establish India as the "AI garage" for the world. However, the opposite end of the digital coin entails the ethical and moral issues associated with open data sharing, primarily associated with patient privacy. The Digital Personal Data Protection Act (DPDPA) of 2023 was legislated by the GOI to address the concerns regarding the same and it has served to enact severe penalties in relation to the data fiduciaries, such as digital public goods, telemedicine apps, online marketplace, government websites, banking apps, and commercial platforms.[28] Several lacunae exist with regard to the Indian medical sector, including legal acceptability of implied consents, roles and responsibilities of data fiduciaries during digital sharing of medical data by the patients, roles and responsibilities of the data processors, and the need for establishment of fortified cybersecurity systems for Patient Healthcare Information (PHI) systems.[29] The Collective Benefit, Authority to Control, Responsibility, and Ethics (CARE) principles for indigenous data governance advocate CARE to ensure the same.[30] The CARE principles are congruent with the holistic data stewardship philosophy of Ayurveda, which emphasizes societal benefit and responsibility.

The creation of unified digital healthcare platforms also serves as large data repositories with PHI of millions at their disposal. Big data analysis utilizing the same can predict disease trends, treatment outcome,[31] perform health technology assessments, and health financial assessments. However, access to such large data for research purposes is fraught with ethical dilemmas and is intertwined with the regulatory frameworks of the countries. MyHealthWay: a Korean healthcare platform designed to facilitate the integration and transfer of medical data from various sources has also reported similar concerns regarding standardization and access to healthcare data for secondary uses such as research.[32] In that regard, DPDPA dictates a general allowance of use of digital data for research through the application of informed consent, whereas the former is through

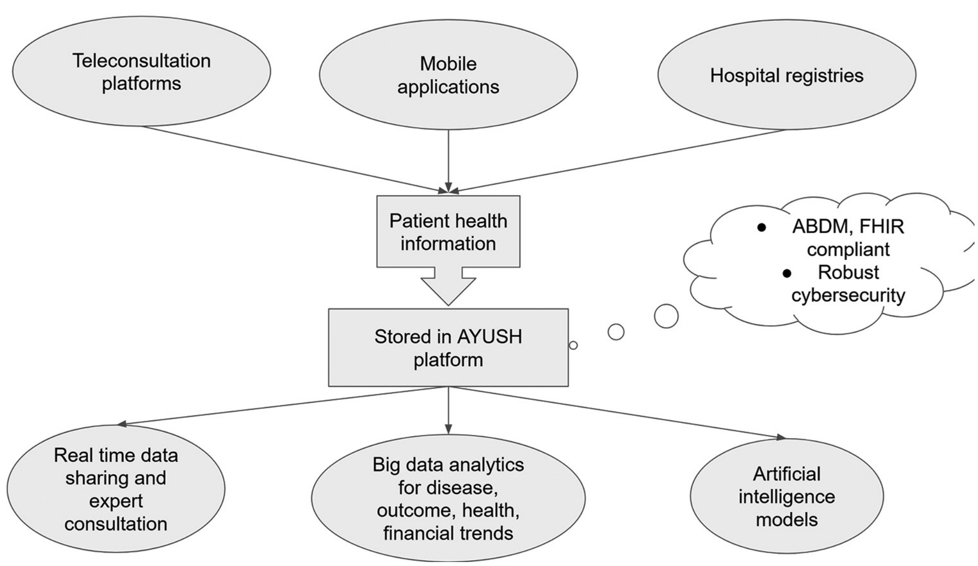

Figure 1: Integration framework for digital Ayurveda ecosystem. ABDM: Ayushman Bharat Digital Mission; FHIR: Fast Healthcare Interoperability Resources

the application of informed consent, and the latter is ambiguous.

Future outlook

While significant inroads have been made for the digital health care to gain ground in the field of Ayush, lacunae exist in the absence of dedicated data centers recording patient outcomes and/or symptoms for developing AI/ML architectures for forecasting clinical outcomes[33] and treatment success.[34] ABDM-compliant digital research repositories wherein researchers host raw research data for government-funded projects need to be considered for the AI/ML leap to take place in a short time. Goal-oriented research projects with field implementation vision and back-end industry support should be encouraged to develop Ayush-based smart technologies or medical devices. Mobile application-based services for the delivery of Ayush medicines need to be promoted to engage the communities and increase awareness regarding the alternative medicines.

The roadmap for the digital Ayurveda ecosystem envisions a phased, data-centric transformation aligned with the ABDM and the National Digital Health Strategy.

Short-term goals (1–2 years)

The immediate focus should be on digitization and standardized data capture across Ayush hospitals and dispensaries. Implementation of A-HMIS should be expanded nationwide, ensuring comprehensive coverage of both public and private Ayush facilities. This phase also involves training healthcare personnel in EHR usage and developing structured templates for Ayurveda-specific clinical data. Suggested measurable indicators: number of Ayush hospitals onboarded on A-HMIS; at least 100,000 patient records digitized; and >80% practitioners trained in digital documentation.

Medium-term goals (3–5 years)

The next phase emphasizes interoperability and data integration with ABDM registries in collaboration with institutions such as the Central Council for Research in Ayurvedic Sciences (CCRAS), Centre for Development of Advanced Computing (CDAC), and other strategic partners within the Ayush research and digital health ecosystem. This includes linking Ayush health facilities (HFR), Ayush practitioners (HPR), and patient ABHA IDs for seamless data exchange. Simultaneously, the development of Natural Language Processing (NLP) tools for Sanskrit and traditional medical terminologies will facilitate the mining of classical texts and clinical narratives. Suggested measurable indicators include the number of Ayush facilities linked to ABDM; number of interoperable datasets made accessible via ABDM sandbox; and 2–3 functional NLP prototypes for Ayurvedic data mining.

Long-term (>5 years)

The final phase envisions a mature ecosystem leveraging AI/ML-driven predictive analytics for individualized Ayurveda health profiling. Suggested measurable indicators include the number of AI/ML models trained on Ayurveda datasets; creation of at least one national Ayurveda digital repository; and evidence of improved patient outcomes via AI-guided decision support.

This three-tier strategy lays a structured, measurable pathway to transform Ayurveda into a digitally empowered, data-driven, and globally interoperable healthcare system.

CONCLUSION

The digital transformation of Ayush demands data-centric integration to enable AI/ML-driven research and clinical forecasting. Establishing ABDM-compliant repositories for patient outcomes and research data is essential. A three-tier roadmap envisions: short-term (1–2 years) digitization through A-HMIS expansion; medium-term (3–5 years) interoperability via ABDM integration and NLP tools in collaboration with CCRAS and CDAC; and long-term (>5 years) predictive analytics and personalized Ayurveda health profiling. Goal-oriented, collaborative projects and mobile health initiatives will strengthen accessibility, innovation, and evidence-based practice, positioning Ayurveda as a digitally empowered and globally interoperable healthcare system.

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Conflicts of interest

There are no conflicts of interest.

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